ME159 Final Project

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1 Introduction

My previous 3D printed door hook initially performed well, but after supporting multiple jackets, it plastically deformed and became unusable. With nowhere to hang my jackets, my room became messier. To address this, I wanted to redesign it using shape optimization to find forming features that could improve stiffness without significantly increasing the material usage. My current 3D printed clothing hook can be seen in Figure 1, where I attached another 3D printed part with a rubber band to increase the strength of the hook. This DIY fix is messy and lazy, so through this optimization, I hope to learn methods to create a stronger hook.

The optimization I will perform focuses on the top portion of the hook, which is the section that interacts with the door. I chose to model this as a shell to keep the geometry simple and computation time low. The goal is to maximize the stiffness of the hook while minimizing deflection, so that the PLA material does not yield under the original failure load. The constraint driving this optimization is total strain. By limiting how much the hook can deform, I can ensure that the redesigned version is strong enough to actually hold my jackets and keep my room clean.

2 Background

For this optimization, I plan to use the Shape Optimization of a Shell module in COMSOL as a guide. In that example, a structural steel L-shaped shell is subjected to a load applied along one edge. Symmetry constraints and fixed boundary conditions are used to restrict motion along the appropriate sides. The optimization then modifies the shell geometry by introducing bends that increase the overall stiffness of the structure.

I intend to follow a similar approach for my own optimization. However, I will modify several parameters so that the model reflects my specific case. In particular, I will change the material properties, loads, constraints, and geometry, while keeping the same optimization framework and solver settings. Using the Shape Optimization of a Shell module as a reference has helped clarify the steps required to properly run my simulation.

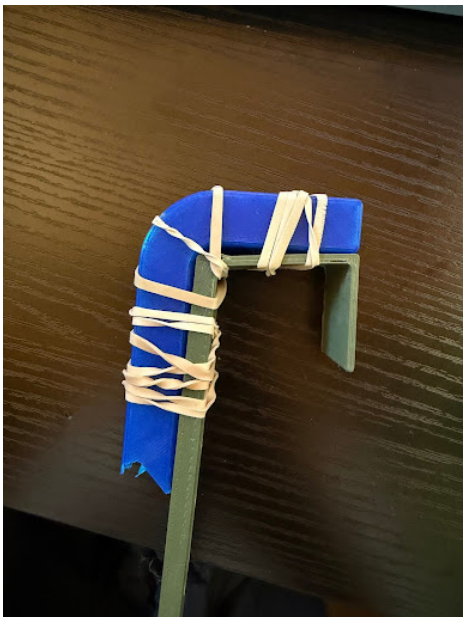


Figure 1: Real Life Hook

3 Method

3.1 Parameters

The simulation uses two key parameters to define the loading condition and allowable deformation:

- **Load:** $F_{load} = 19.208$ N. This value represents the applied force in the simulation and will be discussed in more detail later.
- **Maximum Displacement:** $L_{max} = 0.7$ cm. This limit was chosen as the maximum deformation of the hook during optimization.

3.2 Geometry

The hook geometry is modeled as a 3 cm $\times$ 3 cm square shell with an open top, sized to match my door's width of approximately 3 cm. The sketch was extruded 1.25 cm to represent half the functional length of the hook, since I am using a symmetry constraint. The resulting 2D and 3D geometry can be seen in Figure 2 and 3, respectively.

3.3 Material Properties

The material used is 100 percent PLA infill, which is the filament with which I usually print the door hook. Material properties were sourced from different

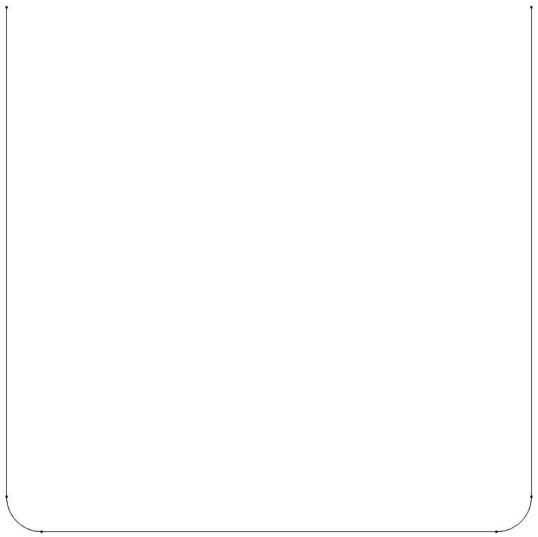


Figure 2: Work Plane Geometry

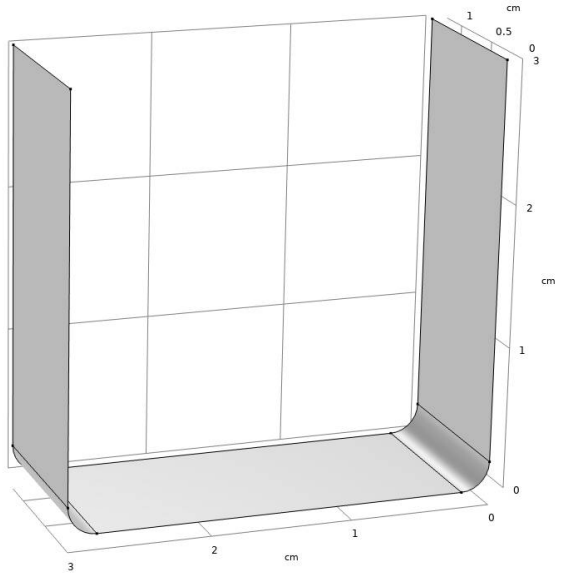


Figure 3: Extruded Geometry

online resources and are an estimated value I manually input into a blank material. The assigned material properties can be seen in Figure 4. For this study, the only required properties were density, Young's modulus, and Poisson's ratio.

Material Contents

Property	Variable	Value	Unit	Property group
Density	rho	1240	kg/m³	Basic
Young's modulus	E	2e9	Pa	Young's modulus and Poisson's ra...
Poisson's ratio	nu	0.33	1	Young's modulus and Poisson's ra...

Figure 4: Material Properties

3.4 Boundary Conditions and Loading

For the applied load, I referenced the weight of one of my Arc'teryx jackets and multiplied it by five to establish a worst-case load the hook should withstand. This load is applied as a normal force on the flat face of the hook, simulating the contact force between the hook and the door. This is a simplification, but it keeps the model manageable. The load can be seen in the Figure 5. Referencing the Arc'teryx website for the jacket I own, I found the jacket to be about 490 g.

Since I am located on Earth, I will use Earth's gravity, to find the weight to be.

$$0.490 * 9.81 = 4.802$$

So with gravity, the weight is 4.802 N, so for 5 jackets, the approximate weight is 4.802*5 = 19.208 N.

For boundary conditions, a fixed edge constraint is applied to one of the long edges of the shell, while the opposite long edge uses a symmetry/roller constraint to simulate the mirrored half of the hook. Symmetry/roller constraints are also applied to the top short edges to allow for proper simulation. The locations of these constraints can be seen in Figure 6 a and b.

3.5 Shell Properties

This section describes the specific shell properties used in the COMSOL simulation.

- **Material Model:** The entire shell is defined as a linear elastic material.
- **Shell Thickness and Offset:** The shell thickness was set to 4 mm to approximate the thickness of the final 3D printed component. An offset was applied so that the top surface of the shell aligns with the edge of the geometry.
- **Initial Conditions:** The initial conditions for the simulation are set to zero, since the system begins from a stationary state.

- **Face Load:** A face load of 19.208 N is applied normal to the main face of the hook that contacts the door. This loading face location is illustrated in Figure 5.

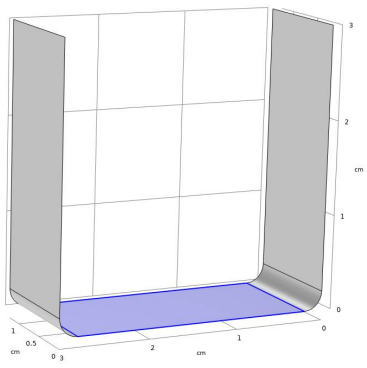


Figure 5: Face Load Location

- **Edge Constraints:**
  - The upper edge of the geometry is defined as a fixed boundary.
  - A symmetry condition is applied along the symmetry plane of the model.
  - The edge opposite the symmetry boundary is defined as a free edge.

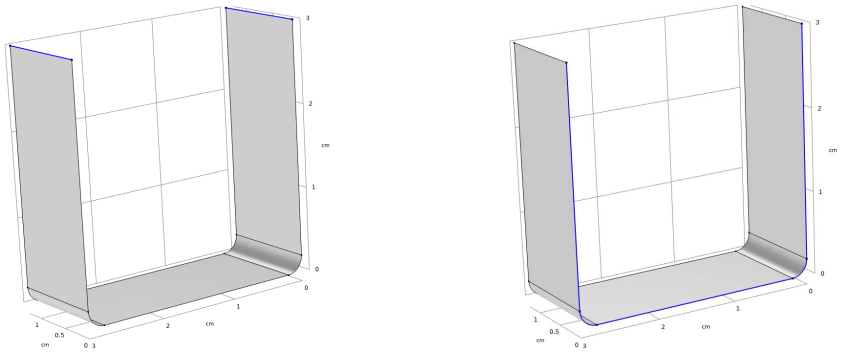
3.6 Mesh settings

For the mesh, I have set the maximum element size at $L_{max}/10 = 0.07$ cm, with a maximum element growth rate: 1.4, a curvature factor of 0.4, and a Resolution of narrow regions of 0.7. The output mesh can be seen in Figure 7. This mesh allows for reasonable results, while keeping computation time reasonable.

3.7 Component Shape optimization settings

For my shape optimization settings, I have similar properties to the Shell settings another previously mentioned. Here, I added a free shape shell feature to all regions. From there I fixed one side, added a symmetry/roller to the opposing edge, and then another symmetry/roller to the two free end edges.

Specifically for the Free Shape Shell, I added a control variable with a maximum displacement: of L_{max}, which is 0.7 cm. For my constrol variables initial



(a) Top Symmetry/Roller Constraint (b) Edge Symmetry/Roller Constraint

Figure 6: 6a illustrates the location of the top Symmetry/Roller edge and 6b illustrates the location of the edge that is mirrored for the whole hook, where the opposing edge has a free edge constraint applied to it. The location of the applied constraints are highlighted in blue.

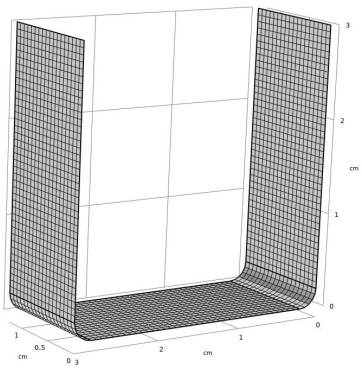


Figure 7: Mesh image

values, I used the equation below.

$$\sin\left(\frac{Y_g - X_g}{1\,\mathrm{m}}\right)\cos\left(\frac{2Z_g}{0.25\,\mathrm{m}}\right)L_{max}$$

Both of these settings follow the Shape module I referenced and did not require changing.

3.8 Study 2 Optimization Settings

For the shape optimization conducted, I used an MMA solver. The rest of the settings are listed below:

- **Method:** MMA
- **Optimality Tolerance:**= 0.001
- **Max number of Iterations:** 50
- **Objective function:** Minimize total elastic strain energy.
- **Control Variables:** Free Shape Shell 1 settings

Using these settings allowed me to get the results for a formed hook, which will allow for a higher stiffness. All of these settings are the same as the reference shape optimization of a shell module.

4 Results

Here are the collection of my results for two different studies, one stationary static study, one shape optimization study, and a stressed shape optimization study.

4.1 Study 1: Stationary

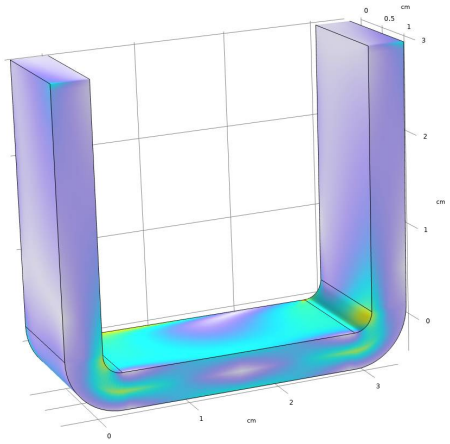


Figure 8: Study 1: Stationary Stress results

First, to conduct a shape optimization, I first wanted to see the initial stress that my shell feels due to the face load. Using a stationary study with the load and boundaries stated above I received the stress results that can be seen in Figure 8. From these results, one can see that there is a built up stress along the hook through the blue areas. Along with that, there are some yellow areas near the fillet that illustrate higher stress regions. This stress distribution isn't beneficial for my design and will overtime cause my clothing hook to slowly deform and break.

4.2 Study 2: Shape Optimization

From there, I conducted a shape optimization using the settings given above, which produced a modified shell after about 15 minutes of computation time. This modified shell can be seen in Figure 9. In the optimized shell, the material forms away from the center near the top two edges, while forming in the opposing region as it moves lower. The bottom edge of the hook also turns into a curved shape with slight bends. These bends move material away from the center of the shell and add features that help increase stiffness against the opposing load.

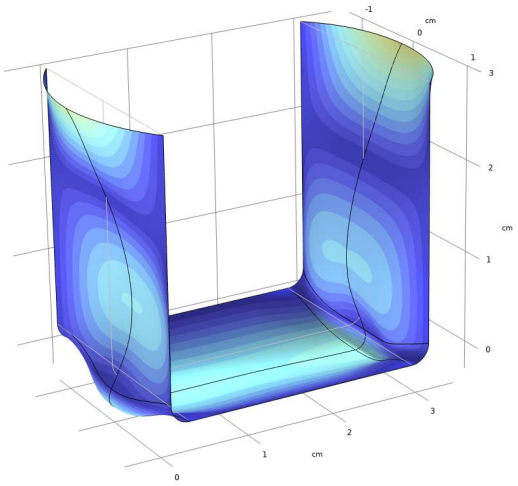


Figure 9: Optimized shell

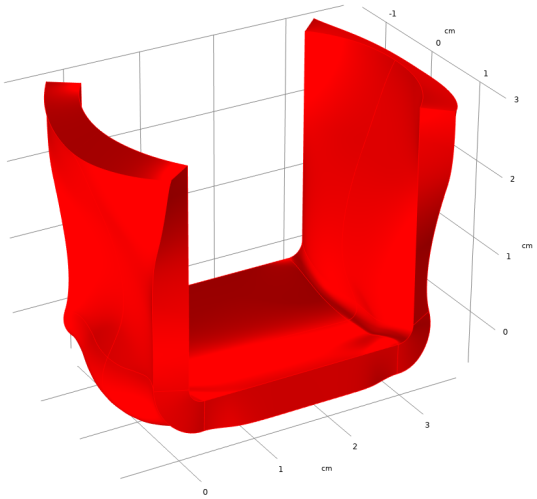


Figure 10: 3D shape Result

When applying the shell thickness to the shape-optimized plane, the resulting 3D geometry can be seen in Figure 10. This image shows the full hook in its 3D form. Note that the thickness is offset so that it begins at the top boundary of the shell.

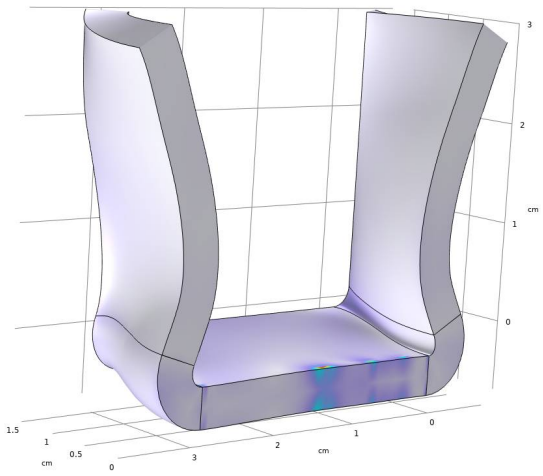


Figure 11: Optimized Shape Stress Result

Figure 11 displays the stationary study ran again on the newly optimized shell. Just from observation, one can see that the stress regions massively decreased, with some hot spots on the symmetry constraint edge. This image gives a cleared image of the bends created, as well as the bends impact on increasing stiffness of material. This result is reassuring and gives me confidence to use these types of forms for my door clothing hook to increase it's strength.

5 Conclusion

After conducting this shape optimization, I came to a few conclusions regarding both my study and COMSOL itself.

Regarding my study, I began to notice the types of patterns that appear in a shell under load that help increase its strength. This realization has made me more aware of similar forms used in sheet metal structures in everyday life that serve similar purposes. I also found a lot of value in the small bends that appear in the optimized geometry. At first glance, it would not seem like this type of bend would add much strength, but after many trials and long computation times, I observed that the stress difference between the original shape and the optimized shape is quite significant.

Through this study, I also gained a broader understanding of one of the phrases Professor Meinhart often says: "COMSOL is an art." While still being relatively new to COMSOL, I have found that running simulations in COMSOL involves many interconnected settings, and even a small change can affect the rest of the model. This experience has given me a greater respect for COMSOL as a tool, as well as a deeper understanding of how to diagnose potential issues in a simulation. Even with a relatively simple simulation like the one described above, I encountered several challenges, but I recognize that working through those difficulties is part of the process.

I plan to take the results of this optimization and use them not only to create a new door clothing hanger, but also to apply the insights I gained to my next COMSOL project.